Data-driven approach to predict customer reviews

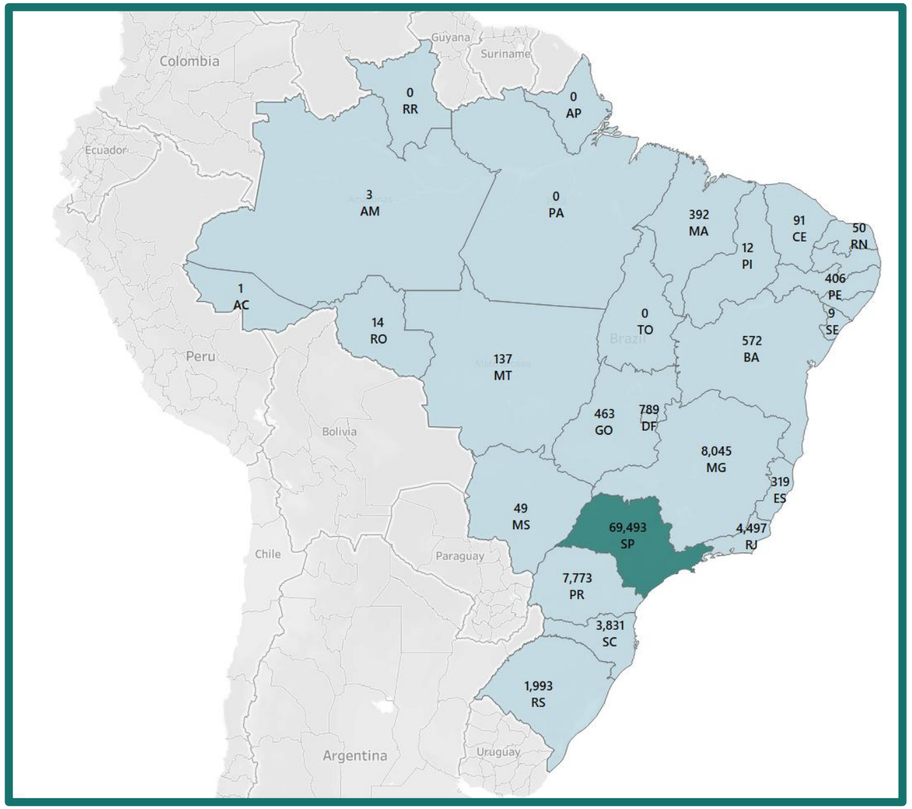
PROJECT SCOPE

The Challenge	Increasing customers positive reviews • Boosting sales • Building customer trust • Attracting new users
Solution	Developing a model to identify customers who are likely to **leave positive reviews.**
Key Advantage	The model not only predicts positive reviews • Minimizing the chance of false positives • Ensuring greater accuracy in targeting the right customers

NILE DATA

Number of Sellers	3,095
Number of Orders	99,441
Number of Customers	96,096
Number of Reviews	89,322

Order Distribution



What might affect the review score?

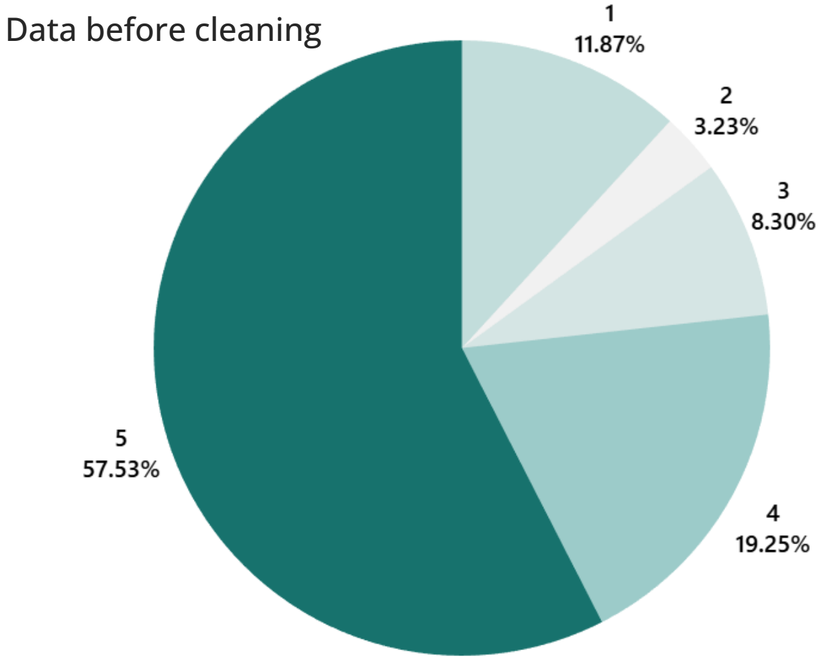
01 Whether the seller and the customer are in the same state

02 Whether the seller and the customer are in the same city

03 Distance between seller and customer

NILE DATA

Number of Reviews	**89,322**
Average Reviews Score	**4.072**



Word Cloud for 4-5 Review Scores

Word Cloud for 1-3 Review Scores

Delivery time may impact customer ratings.

NILE DATA

Time Related Data

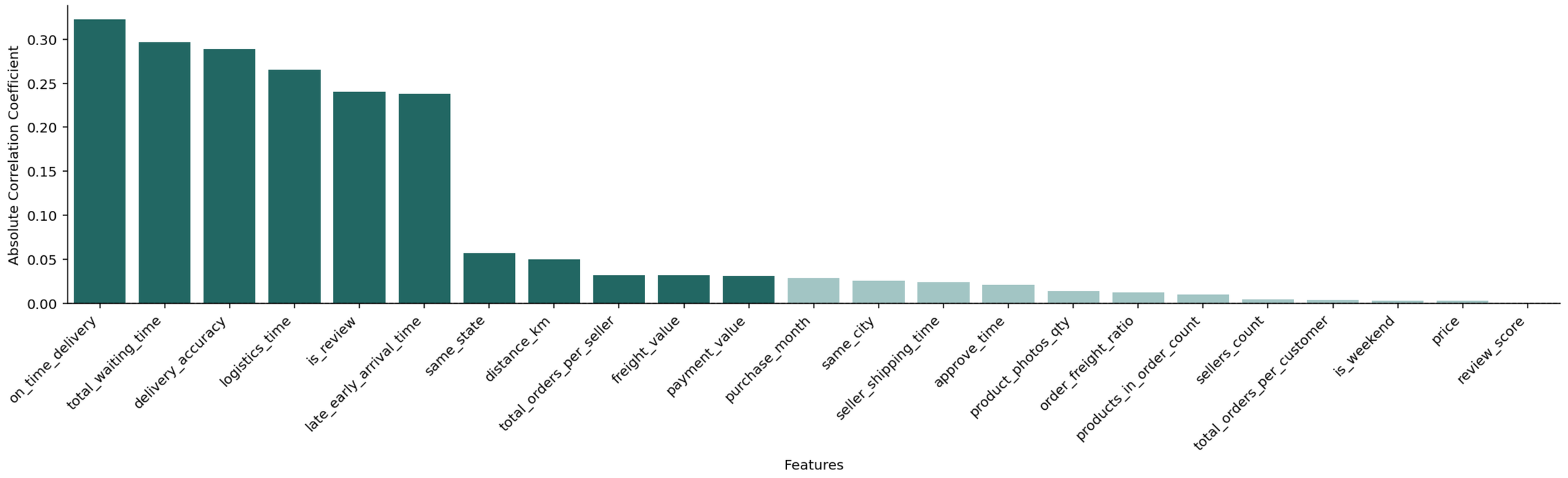
1	Order Purchase Date	The date that customer makes order
2	Order Approved Date	The date that payment was approved
3	Order Shipping Date	The date that seller sent to logistic
4	Carrier Delivery Date	The date that the customer received the order
5	Order Estimated Delivered Date	The date that the platform estimated the customer would receive the order

What might affect the review score ?

Overall Side	Estimated vs. Actual Delivery Time Difference
	Total waiting time
Platform Side	Approve time
	Estimated Delivered Date accuracy
Seller Side	Seller shipping time
Logistic Side	Logistics time

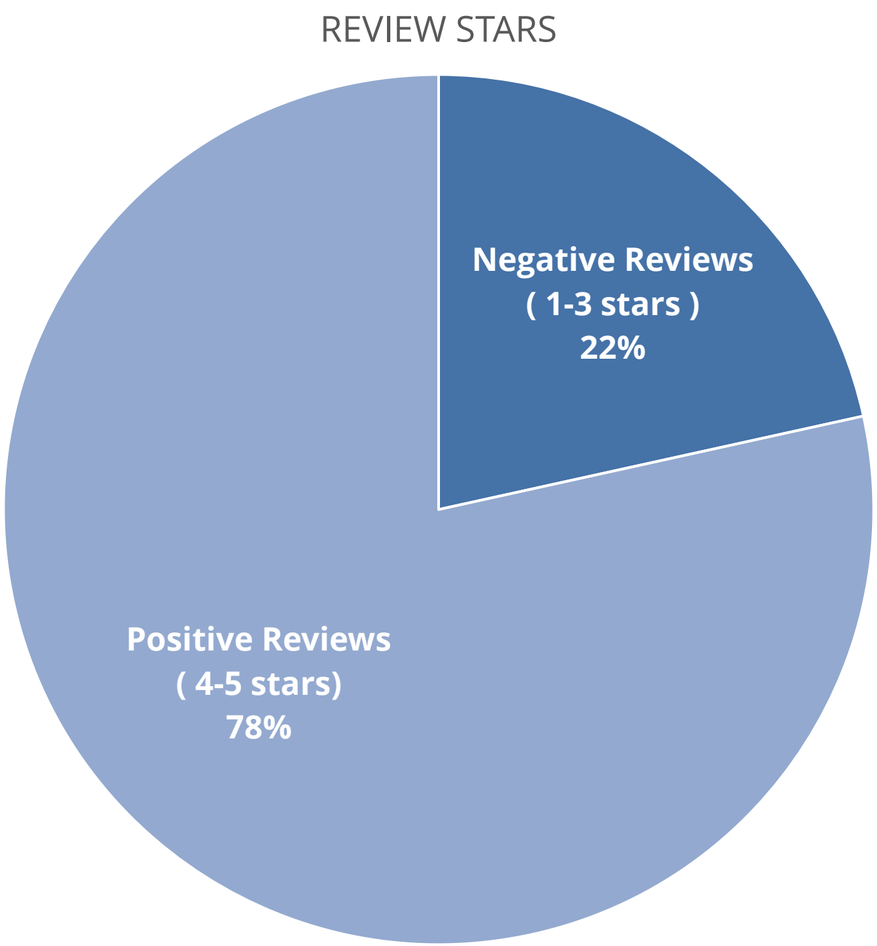
WHICH FACTORS INFLUENCE CUSTOMER REVIEW THE MOST?

Absolute Feature Correlation with Review Score



Retain the top correlated features with the review score for model training.

MODELLING



Data source : Based on cleaned data, not the original dataset

Classification

Goal: Predict which customers are likely to give a positive review (4-5 stars)

Review star classification :

- Positive review: 4-5 stars
- Negative review: 1-3 stars

Benefit: Improved model focus (higher accuracy and precision)

Solution for class imbalance

Challenge: The dataset was imbalanced, with **predominantly positive reviews**, which could bias the model.

Solution Tested:

- Class Weight Adjustment
- SMOTE (Synthetic Minority Over-sampling Technique)

Machine Learning Algorithms

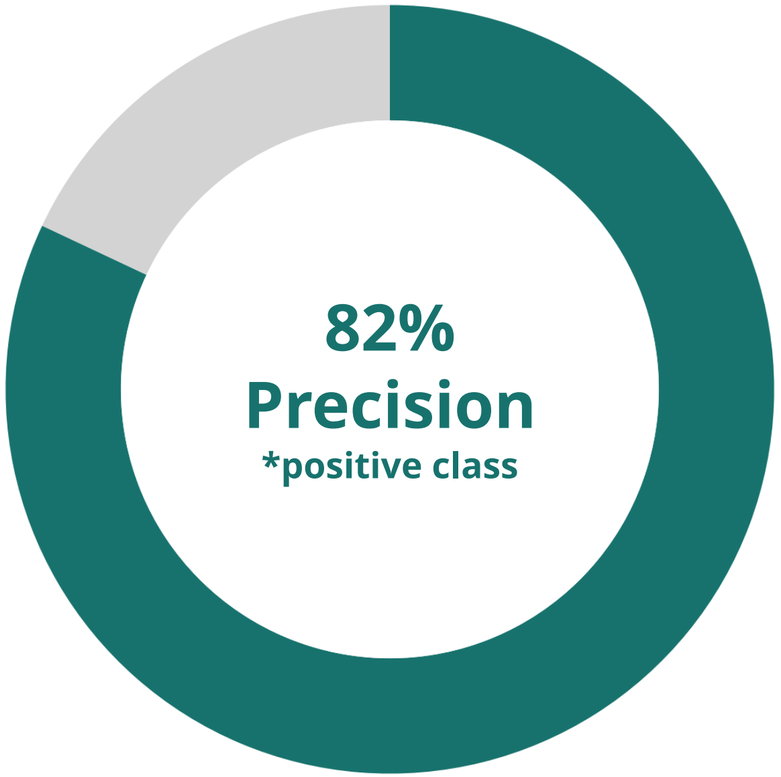
Random Forest, Gradient Boosting and XGBoost

MEASURING SUCCESS: HOW OUR MODEL DELIVERS RESULTS

Key Criteria for choosing the best model

Overall Reliability	Measures how often the model's predictions are correct
Precision	Focuses on how many of the positive review predictions were truly correct
Recall	Checks how well the model captures all the positive reviews
Balanced Performance	Provides a balanced view of both precision and recall, ensuring consistency

HOW OUR MODEL STANDS OUT



Model

- **Algorithms:** Random Forest
- **Class Imbalance Techniques:** Class Weight Adjustment
- **Model precision score:** 82%

Overall Performance

- **High Performance in Predicting** whether customers will leave positive or negative reviews.
- **High Precision for Positive Reviews**
  - High precision in predicting positive reviews
  - Minimizing the risk of misclassifying negative reviews.